Smart Shannon: Empowering Communities with Real-Time River & Flood Insights

Vision:

A connected, resilient Shannon River Basin where communities, researchers, and policymakers can monitor water quality, predict flooding, and take sustainable action in real time.

Background:

The Shannon River Basin is vital to Ireland's environment, communities, and economy, supporting millions of people, extensive agriculture, and rich biodiversity. Yet, climate change, pollution, and increasingly severe floods threaten its health.

Traditional monitoring is slow and resource-intensive, leaving communities and authorities reactive rather than proactive. Smart Shannon envisions a data-driven, community-powered river system that delivers real-time insights and predictive intelligence.

Problem Statement:

How might we develop a real-time, open, and community-powered system that continuously monitors river health and predicts flood events, inviting citizens and authorities to respond faster and more effectively?

Key challenges:

Real-time monitoring of water quality and river conditions

Predicting flood events using AI and historical data

Engaging communities in participatory science and reporting

Expected Outcomes

- Technology & Innovation
- Community & Engagement
- Citizen science app for real-time reporting of observations
- Community workshops and hackathons to build local capacity
- Research & Policy
- Open datasets for future research, policy development, and environmental education
- Optional KPIs: Flood prediction accuracy, number of engaged community participants, volume of open data shared

Scope

Included:

River and flood data collection (sensors, satellite, open datasets)

Predictive analytics, mapping, and visualization

Community engagement and open data-sharing tools

Excluded:

Large-scale engineering works or infrastructure changes

Proprietary or closed-source solutions

Difficulty:

Intermediate to advanced. Requires a lot multidisciplinary collaboration across different fields (e.g Environmental science, IoT and hardware prototyping, Data analytics and AI, Design and community engagement)

Resources:

Data: EPA Ireland, Met Éireann, Copernicus, local authority datasets
Knowledge & Expertise: TUS RISE partners, environmental scientists, local councils
Tools & Tech: Arduino/Raspberry Pi kits, GIS software, cloud dashboards
Spaces & Community: Shannon Studio labs, hackathons, workshops
Tensorflow and Pytorch(Executorch) IoT guides and kits. hese are useful for prototyping and creating a machine learning system that is accurate and fast.
Potential use of drones and image recognition to assess water quality visually.
Machine learning combined with low cost IoT sensors for water quality and flood levels
to analyse trends and predict flooding and pollution.

Inspiration:

Safecast: Radiation and environmental monitoring by volunteers after disasters
FixMyStreet / SeeClickFix: Community reporting platforms that can inspire flood or pollution reporting

Thames21 (UK): Because it is a community-led water monitoring network.

Smart Dublin: Provides a lot of open data for urban challenges.

Copernicus Open Access Hub: Is very useful for satellite imagery used in environmental analysis.

Global Water Watch: Combining AI and satellite data to track water changes

Through Smart Shannon people may adapt these models to create a uniquely Irish, community-focused river monitoring ecosystem to ensure that the Shannon river will be able to support our valuable industries and our way of life in the future